

Generalization and data splits

The point of learning is useful behavior on future data, not remembering the examples used to adjust the model.

15–20 MINUTES

Prerequisite: optimization

SPLIT-DESIGN EXERCISE

1 • Fit Performance on training examples.	2 • Generalize Performance on new examples.	3 • Split Separate learning and measurement.	4 • Shift Deployment differs from sampled data.
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Fitting is necessary but insufficient

A model must fit useful patterns in the training set. It can also fit accidental details. **Overfitting** means the model performs much better on training data than on relevant unseen data. **Underfitting** means it has not captured enough of the useful pattern even on training data.

Central mental model

Training data changes the model; validation data guides choices; test data estimates final performance after choices are fixed.

Three different jobs

Split	Who may use it?	Purpose
Train	Optimizer	Update model parameters
Validation	Developer	Choose settings and checkpoints
Test	Final evaluator	Estimate performance after choices

Repeatedly inspecting test failures and changing the model turns the test set into another validation set. A new untouched test is then needed for an honest final estimate.

Leakage and correlated frames

Data leakage lets information from evaluation influence training or selection. In video, randomly splitting neighboring frames is especially risky: adjacent frames can be nearly identical. The validation set may look “unseen” by filename while sharing the same scene, lighting, camera position, and cube pose with training.

A better split often groups by capture session, physical setup, object instance, or time. The grouping unit should match the kind of novelty expected in deployment.

Browser exercise: split a video dataset

You recorded three 60-second videos: desk daylight, desk evening, and robot room. Which split better tests the robot-room deployment?

A

Shuffle every frame, then split 80/10/10.

B

Keep capture sessions together; hold out robot-room sequences.

Reveal the reasoning

Distribution shift

Training and deployment examples come from distributions: recurring combinations of scenes, cameras, lighting, viewpoints, and objects. **Distribution shift** means these combinations differ. Evaluation should sample the conditions the system must handle, including hard negatives and rare but important cases.

Applied to our cubes

MEASURED EVIDENCE The original documented validation split is only nine images with 21 instances. Strong numbers on that sample have high uncertainty and do not establish performance in the JetRover room.

MEASURED EVIDENCE The current milestone status is M5 PARTIAL because the node runs while reliable live positive detection remains unresolved.

DECISION RULE A future dataset should separate capture sessions and reserve representative robot-room positives and negatives for evaluation.

Project anchors: model notes, milestones, and M5 report.

Check your understanding

Why keep validation and test data separate?

- ☐ To separate selection from estimation
- ☐ To double every training example
- ☐ To remove all difficult images

Explain it back

Explain why randomly splitting frames from one video can give misleading evidence. Propose a stronger split for this project and name the deployment condition it tests.

Primary sources

- Deep Learning, Chapter 5 — capacity, overfitting, validation, and estimators.
- D2L: Generalization in classification — generalization and evaluation.